Name: ______________________________

Class: ______________________________

UNIT EXAM

Unit 3 Gravity and electromagnetism

Time permitted: 70 minutes

	Section	Number of questions	Marks available	Marks achieved
A	Multiple choice	30	30	
B	Short answer	11	40	
	Total		**70**	

Grade: ____________________

Scale:

A+	66–70	A	60–65	B	50–59	C	40–49	D	35–39	E	21–34	UG	0–20

Comments:

For all questions:

- $g = 9.8\ \text{m s}^{-2}$ on the surface of Earth
- The universal gravitational constant $G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
- The charge on an electron is $-1.6 \times 10^{-19}\ \text{C}$
- Mass of an electron or positron is $9.11 \times 10^{-31}\ \text{kg}$
- Mass of a proton is $1.67 \times 10^{-27}\ \text{kg}$
- The permittivity of free space is $\varepsilon_0 = 8.9 \times 10^{-12}\ \text{C N}^{-1}\ \text{m}^{-2}$
- The coulomb constant is $\dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$
- The permeability of free space is $\mu_0 = 4\pi \times 10^{-7}\ \text{T m A}^{-1}$
- The speed of light is $3.0 \times 10^8\ \text{m s}^{-1}$

Section A Multiple choice (30 marks)

Section A consists of 30 questions, each worth one mark. Each question has only one correct answer. Circle the correct answer. Attempt all questions. Marks will not be deducted for incorrect answers. You are advised to spend no more than 30 minutes on this section.

1 A rock is dropped from a stationary helicopter. How fast will the rock be moving 4 s later?

A $19.6\ \text{m s}^{-1}$

B $39.2\ \text{m s}^{-1}$

C $46.5\ \text{m s}^{-1}$

D $78.4\ \text{m s}^{-1}$

2 A cannonball is dropped from a stationary helicopter. Its downwards speed is measured at $38\ \text{m s}^{-1}$ from a platform 110 m high. At what speed will it hit the ground? Ignore air resistance.

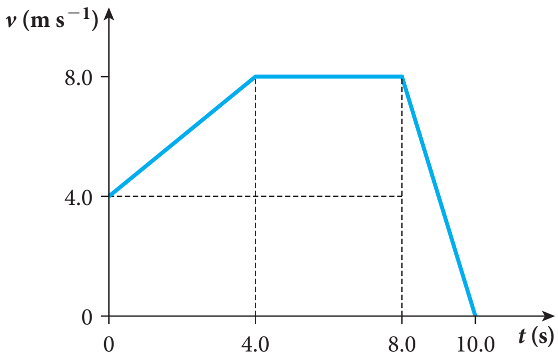
A $27\ \text{m s}^{-1}$

B $49\ \text{m s}^{-1}$

C $50\ \text{m s}^{-1}$

D $60\ \text{m s}^{-1}$

3 Find the net displacement for the particle's motion in the velocity-time graph below.



A –4 m

B 12 m

C 48 m

D 64 m

Use the information below to answer Questions **4** and **5**.

A 400 g weight at the end of a 75 cm string is being whirled in a vertical circle, going around 75 times every minute.

4 What is the period of this motion?

A 0.80 s

B 0.90 s

C 1.20 s

D 1.25 s

5 Find the tension in the string when the weight is at the lowest point in the motion.

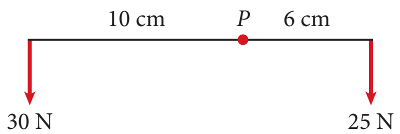
A 3.9 N

B 19 N

C 22 N

D 45 N

6 Find the magnitude of the torque about the point *P* below.



A 20 N m

B 150 N m

C 225 N m

D 450 N m

7 How much work is done by a 400 N force, which moves a 2 kg brick a distance of 25 cm, if the force is acting at an angle of 60° to the direction of the displacement?

A 50 J

B 87 J

C 100 J

D 173 J

8 Find the change in gravitational potential energy when a 12 kg mass falls 2.0 m.

A −24 J

B 24 J

C −235 J

D 235 J

9 A woman has a mass of 65 kg. If she were to weigh herself while in a lift accelerating upwards at 1.96 m s^{-2}, what would the scales show?

A 52 kg

B 65 kg

C 70 kg

D 78 kg

10 What is the gravitational field strength on the surface of Mars, which has a radius of 3400 km and a mass of 6.4×10^{23} kg?

A 3.7 N kg^{-1}

B 4.5 N kg^{-1}

C 6.8 N kg^{-1}

D 9.8 N kg^{-1}

11 What force is an 80 kg man exerting on the planet Mars (see above) as he sits at home, while the planet is 85 million kilometers away from Earth?

A 0 N

B 3.8×10^{-10} N

C 4.7×10^{-7} N

D 1.3×10^{-5} N

12 Find the escape velocity for a particle on the surface of Vesta, of mass 2.59×10^{20} kg and diameter 525 km.

A 181 m s^{-1}

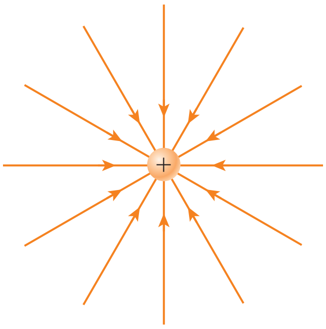
B 257 m s^{-1}

C 32 900 m s^{-1}

D 65 800 m s^{-1}

13 Which of the following shows the electric field around a positive charge?

A



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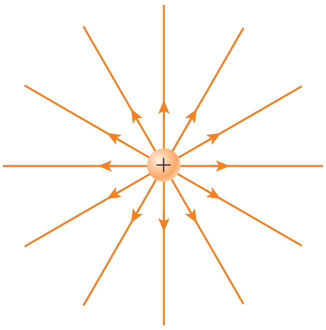
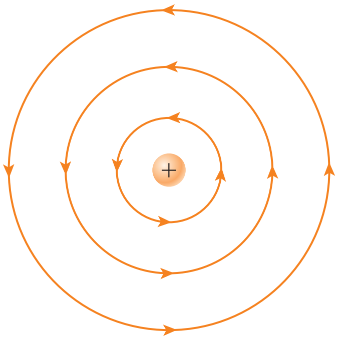
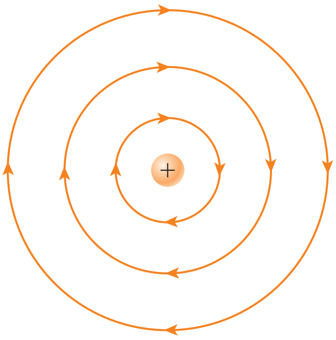


C



D



14 How does the electrostatic force change if the distance between two forces is halved?

A It is also halved.

B It will be one quarter the size.

C It will be doubled.

D It will be quadrupled (four times).

15 Identify the electric field 4.0 m away from a charge of 0.016 C.

A $2.7 \times 10^{6}\ \mathrm{N\ C^{-1}}$

B $4.5 \times 10^{6}\ \mathrm{N\ C^{-1}}$

C $9.0 \times 10^{6}\ \mathrm{N\ C^{-1}}$

D $3.6 \times 10^{7}\ \mathrm{N\ C^{-1}}$

16 Identify the force on an electron from an alpha particle which is 2.5×10^{-10} m away.

A 7.4×10^{-9} N

B 2.7×10^{-10} N

C 5.4×10^{-10} N

D 3.7×10^{-11} N

17 A positron moves 21.0 cm perpendicular to a uniform electric field of 4.5 N C^{-1}. How much work is done by the field?

A 0 J

B 0.42 J

C 0.95 J

D 1.33 J

18 Find the energy in electron volts when a ferric ion, Fe^{3+}, is accelerated through 30 V.

A 10 eV

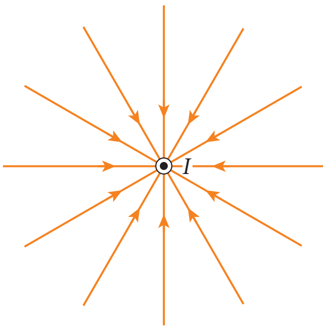
B 30 eV

C 60 eV

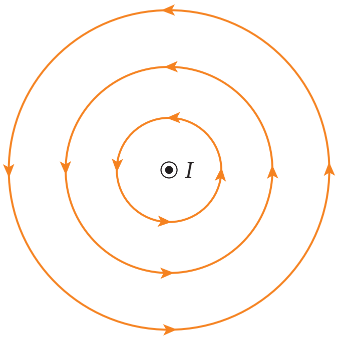
D 90 eV

19 Which of the following represents magnetic field lines, if the current is coming out of the page?

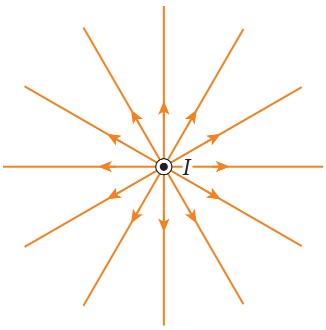
A



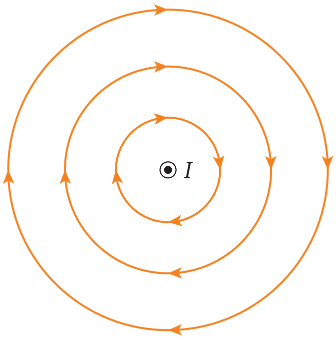
C



B



D



20 How much work is done by a 0.12 T magnetic field if a 1.0 C charge enters the field and moves a total distance of 10 cm?

A 0 J

B 0.012 J

C 0.056 J

D 0.12 J

21 Find the magnetic field strength 15 cm from a long wire carrying a current of 13.5 A.

A 9 μT

B 15 μT

C 18 μT

D 45 μT

22 If the magnetic field strength 60 cm from a long current carrying wire is 90 μT, what is the magnetic strength 20 cm from the wire?

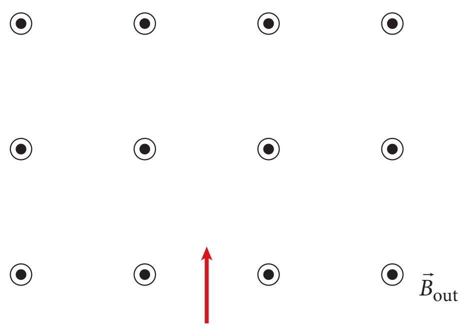
A 10 μT

B 30 μT

C 270 μT

D 810 μT

23 An alpha particle enters a uniform magnetic field as shown below. Which is the correct description of its path?



A Moving left in a circular arc

B Moving right in a circular arc

C Moving left in a parabolic arc

D Moving right in a parabolic arc

24 Which of the following is *not* true of Earth's magnetic field?

A It causes the aurora australis.

B The field lines run from Earth's north magnetic pole to Earth's south magnetic pole.

C It protects us from cosmic rays.

D Earth's magnetic poles are changing location.

25 Calculate the magnetic flux if the vector representing an area of 1.25 m^2 is parallel to a uniform magnetic field of 2.4 mT.

A 0 mWb

B 3.0 mWb

C 5.2 mWb

D 6.0 mWb

26 What is the magnetic flux if the vector representing an area of $1.25\,m^2$ is perpendicular to a uniform magnetic field of 2.4 mT?

A 0 mWb

B 3.0 mWb

C 5.2 mWb

D 6.0 mWb

27 A circular loop of cross-sectional area $0.25\,m^2$ is perpendicular to a magnetic field, which decreases uniformly from 0.10 T to 0 T over 5.0 s. How does the emf change during this time?

A There is no emf produced.

B The emf uniformly decreases from 5 mV to 0 mV.

C The emf uniformly increases from 0 mV to 5 mV.

D The emf remains constant at 5 mV.

28 An AC current has a peak potential difference of 120 V. What is the root mean square (rms) value?

A 85 V

B 96 V

C 170 V

D 240 V

29 A voltage transformer is designed so that the primary coil has 150 turns and the secondary coil has 450 turns. The primary emf is 240 V. What is the secondary emf?

A 80 V

B 160 V

C 480 V

D 720 V

30 Find the wavelength of a radio wave of frequency 774 kHz.

A 2.58 mm

B 69.1 cm

C 388 m

D 3.9 km

Section B Short answer (40 marks)

Section B consists of 11 questions. Write your answers in the space provided. You are advised to spend 40 minutes on this section.

1 A cannon ball is fired at 36 m s^{-1} from the (horizontal) ground at an angle of 40° to the horizontal. How far will it travel before hitting the ground again?

(3 marks)

2 A road around a bend with radius of curvature 80 m is banked at 12° to the horizontal. A car of mass 800 kg travels around the corner at 16 m s^{-1}. Find:

a the magnitude of the centripetal force

b the magnitude and direction of the force on the car by the road, parallel to the plane

c the magnitude of the normal reaction.

(2 + 2 + 1 = 5 marks)

3 a If kinetic energy can be given as $\frac{1}{2}mv^2$, describe the units of kinetic energy in terms of mass, length and time.

b Show that the units for potential energy are the same as for kinetic energy.

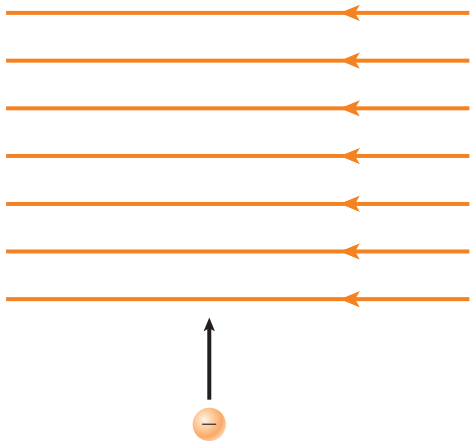
(2 + 1 = 3 marks)

4 a Captain Kirkoff wants to calculate the mass of an unknown planet, so he puts his spaceship in orbit at a distance of 25 000 km from the centre of the planet. If he orbits the planet once every 40 hours, find the mass of the planet.

b To check his result, Kirkoff lands his ship on the surface of the planet, which is almost exactly 700 km in diameter. He drops a lead weight from a height of 10 m to the surface. How long should it take to hit the ground?

(3 + 3 = 6 marks)

5 **a** Find the magnitude and direction of the acceleration of a chlorine ion Cl^- of mass 5.81×10^{-23} as it enters the electric field of 2.5 N C^{-1} as shown below.



b What will be the shape of the path of the ion?

(3 + 1 = 4 marks)

6 A 2.5 g ball bearing has a 2.4 mC charge. It rolls along a smooth flat surface moving at 2.0 m s^{-1}. When it enters a uniform electric field parallel to its direction of movement it is accelerated to 8.0 m s^{-2}.

a Find the work done by the field on the ball bearing.

b Find the change in electrical potential energy.

c Find the potential difference during the balls motion through the field.

(1 + 1 + 1 = 3 marks)

7 A wire that carries a 4.0 A current makes an angle of 30° with Earth's magnetic field, which is 100 μT at that point. What force per unit length does the wire experience?

(2 marks)

8 **a** Find the force on, and the acceleration of an alpha particle, if its speed while entering a magnetic field of 2 T at right angles is 400 km s^{-1}.

The mass of an alpha particle is 6.64×10^{-27} kg.

b Find the radius of the subsequent path of the alpha particle.

(2 + 1 = 3 marks)

9 Two parallel wires 2 mm apart and 50 cm long carry currents of 4.0 A and 2.8 A in the same direction. Use the strength of the magnetic field at the 2.8 A wire from the 4.0 A wire to find the magnitude and direction of the forces on both wires.

(3 marks)

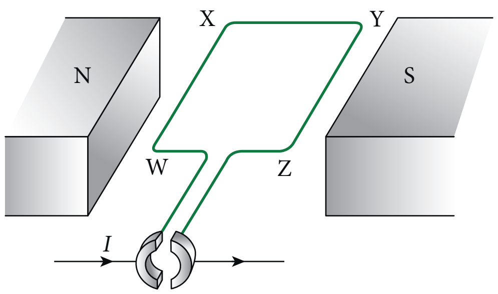
10 A 90 cm $\times$ 40 cm rectangular coil is made up of 500 turns. It initially sits perpendicular to a magnetic field of 0.15 T, and is rotated 5 times a second.

a Find the maximum emf produced.

b Sketch the emf as a function of time, showing at least two cycles.

(1 + 3 = 4 marks)

11 The diagram below shows a loop of wire in a uniform magnetic field. The current flowing ($I = 2.4$ A) is constant in the direction shown. The magnetic flux is a uniform 0.05 T. WX = ZY = 20 cm and XY = 15 cm.



a Find the magnitude and direction of the force on:

i WX

ii XY

b What is the moment produced?

c Ignoring friction, what would happen if the ring at the front wasn't split?

(1 + 1 + 1 + 1 = 4 marks)